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Research Article

Emotion-Aware Indian Sign Language Recognition: A Multimodal Approach With Sign-Expression Correlation Analysis

Hriday Ranka[1], Darshit Sarda[1], Haardhik Kunder[1], Aaditya Rajesh[1], Aniket Kore[1]

[1]Department of Computer Engineering, Dwarkadas J. Sanghvi College Of Engineering, Mumbai, India.
hridayr1234@gmail.com; darshitsarda10@gmail.com; kunderhaardhik@gmail.com;
aadityarajesh04@gmail.com; aniket.kore@djsce.ac.in;

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ARTICLE INFO **ABSTRACT**

This research presents a novel approach in developing a real-time sign language recognition system for both Indian Sign Language (ISL) and the emotions and feelings of the signer, aiming to assist those with hearing impairments and those who cannot understand sign language. The system utilizes CNN, Long ShortTerm Memory (LSTM) model and NLP techniques to ensure efficient and effective interpretation of signs and emotions. This dual-purpose system reduces the need for human translators, making it versatile for use in hospitals, schools, courts, training institutions, and various other settings. The combination of hand sign recognition with facial expression analysis enhances communication by providing more natural interactions and offering additional context about the signer's emotional state. The emotion detection component employs advanced computer models to analyze facial expressions, thereby improving the understanding of the signer's feelings during communication. A key innovation of this research is the creation of a custom dataset for Indian Sign Language, rather than relying on existing datasets. Additionally, the integration of sign language and emotion detection into a single system, coupled with correlation analysis, distinguishes this project as unique. Furthermore, the system includes a feature that converts written words into spoken words using Google Text-to-Speech in 5 different Indian regional language, thereby facilitating easier communication for all parties involved.

Keywords: LSTM, CNN, Indian Sign Language(ISL), Correlation Analysis

1 Introduction

In the sphere of computer assisted communication to individuals, the trend is to incorporate the elements of machine learning and sign language reading in addition to other activities to achieve greater inclusivity and communication of persons with hearing problems.Attesting to the inclusion of sign language as a regular subject in schools is significant since it will benefit individuals who have hearing problems. The aim for a fine recognition system of sign language is due to the difficulties that the deaf people have to deal with in their social life. This system will be doing away with the tapping-on-time, hassle of learning sign-language and interpretation services in various fields including medical, schools, training sessions etc. This technology reduces the requirement for an ISL interpreter because it recognises ISL signs as well as emotions.

The current research puts forth a new system which is capable of showing real-time Indian Sign Language (ISL) interpretation and emotion recognition in one combined digital aid. This is a worthy investment of work because the need for effective communication tools has arisen. The integration between sign language interpretation and emotion recognition shows the development of new breakthrough in assisted communication mode, which has the potential to allow the users to convey their emotions and language at the same time.

While passing through the dynamic terrain of assistive technologies this solution has the capacity for overcoming the existing communication flaws. The system to be developed is intended to flout the conventional assistive communication paradigm by seamlessly integrating an intelligent technology with the intricate language expression of humans. This method is designed to bridge the gap that humans commonly face when communicating with the impaired by promoting inclusivity and accessibility. The advantage of this technique

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over others is that it is not confined to the limits of routine systems and is aimed at giving the impaired community a tool for deciphering the ISL gestures accompanied by the emotional complexities.
This research goes beyond the sphere of technology, depicting even more the meaning of our initiative to establish a more united and connected planet. Moreover, in this research, a novel idea is to integrate Google Text-to-Speech[1][2] service that transforms written text to voice notes. It helps the system to acquire more powerful communication skills.

2 Literature Review

In recent years, the advancement of Facial Emotion Recognition (FER) techniques has been fueled by the widespread adoption of Convolutional Neural Networks (CNNs). [3] Achieving an impressive 93% accuracy in real-time, it exceeded previous standards in the field and has reached an impressive milestone. Over the last few years, FER has benefited greatly from the incorporation of CNNs as a common approach in many fields. According to the research done by [3] at the SMART–2021 conference, FER plays an important function in microsocial interactions and technologies. It received 93% accuracy in real-time, which set a new benchmark for the previous methods in the field and was an achievement. Based on such datasets as LFW and The Extended Yale Face Database B, the study suggested the incorporation of face processing technology for the enhancement of FER, which led to an interesting direction for future research and development.
Likewise, [4] also provided an extensive review of real-time facial emotion detection and classification with the help of numerous deep learning models such as CNNs, DL, and Haar Cascade[5]. They proposed their classification model which, using the Fer2013 dataset, obtained 97% of accuracy on seven different classes. Histogram equalization and background subtraction techniques were integrated which enhanced the classification accuracy and provided a basis for a precise real-world FER algorithm.
On the other hand, there has been growing attention and studies on the sign language recognition especially in the Indian environment. The paper of [6] proposes a CNN-based approach for Indian Sign Language (ISL) recognition based on the visual signals with the purpose of breaking the communication barrier by converting text/audio using Canny Edge Filtering and text to audio conversion techniques with an accuracy of 95.31% accuracy rate. Besides, their study also focused on the issues related to gesture recognition in ISL and also highlighted the need of integrating audio resources for the provision of better communication facilities for the hearing impaired people which in fact helps in creating other modes of communication. Their study not only addressed the challenges of ISL gesture recognition but also highlighted the importance of integrating audio resources to improve the achievement of individuals with hearing loss, and thus have also emphasized the facilitation of inclusive communication channels.
[7] further advanced the field by exploring real-time ISL using deep learning techniques. CNNs, Media Pipe, and OpenCV were used to develop an efficient system for finding and distributing ISL gestures in real time, effectively reducing the communication gap between Deaf and hearing individuals. Their findings underlined the effectiveness of CNN multimedia programs in improving communication and inclusion with the Deaf community, highlighting the adaptability of such technologies in enabling low-density communication.
Additionally, [8] focused on expanding American sign language (ASL) recognition and conversion into text in real time, along with emotion recognition. Using a combination of CNN, SVM, HAAR, Kmeans, LSTM algorithms, they achieved a model accuracy suitable for around 80% Their research highlighted the importance of dynamic hand signatures and emotion recognition to reduce communication barriers between sign language users and non-signers, and promoted accessibility and inclusion of hearing-impaired individuals.
This literature review, highlights the significant advancements in sign language recognition and emotion recognition using machine learning applications. All of these studies advocate the inclusion of sensory and audio components in sign language recognition systems, allowing hearing-impaired community members to step up and be touched, and pave the way for touching interactive media more about everyone in social situations.

3 Proposed Methodology

Our innovative approach offers an all-inclusive solution to enhance communication accessibility for people with hearing impairments who use sign language with special attention to the Indian Deaf community . The system consists of four key components, a hand sign recognition module, an emotion recognition module, a text-to-speech converter supporting five regional Indian languages and a correlation NLP module.

3.1 Palm Detection Model using MediaPipe

The Palm Detection model uses a pre-trained MediaPipe pipeline to recognize and track hand symbols in a video stream. Initially, the code captures frames from the video feed (30 frames) and processes them for hand detection. MediaPipe's hand tracking model then detects and tracks the keypoints of the palm, fingers, and wrist in each frame. The detected keypoints, organized into a NumPy array, serves as input for further analysis. Every keypoints position is normalized and converted to floatingpoint arrays using NumPy. LSTM model is trained on these keypoints and it learns to recognize and interpret complex hand signs. For each 30 frames in the video stream this step is repeated providing real-time feedback on the hand signs being performed by

individuals in the video. When no hands are detected a corresponding message is displayed on the output screen which ensures continuous monitoring and interpretation of hand gestures.

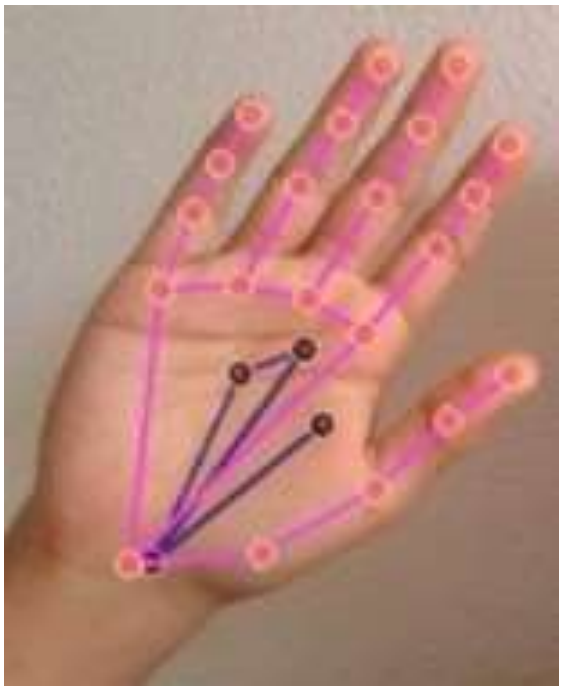

Fig. 1 Identifying all the hand landmarks in different palm configurations such as open palm

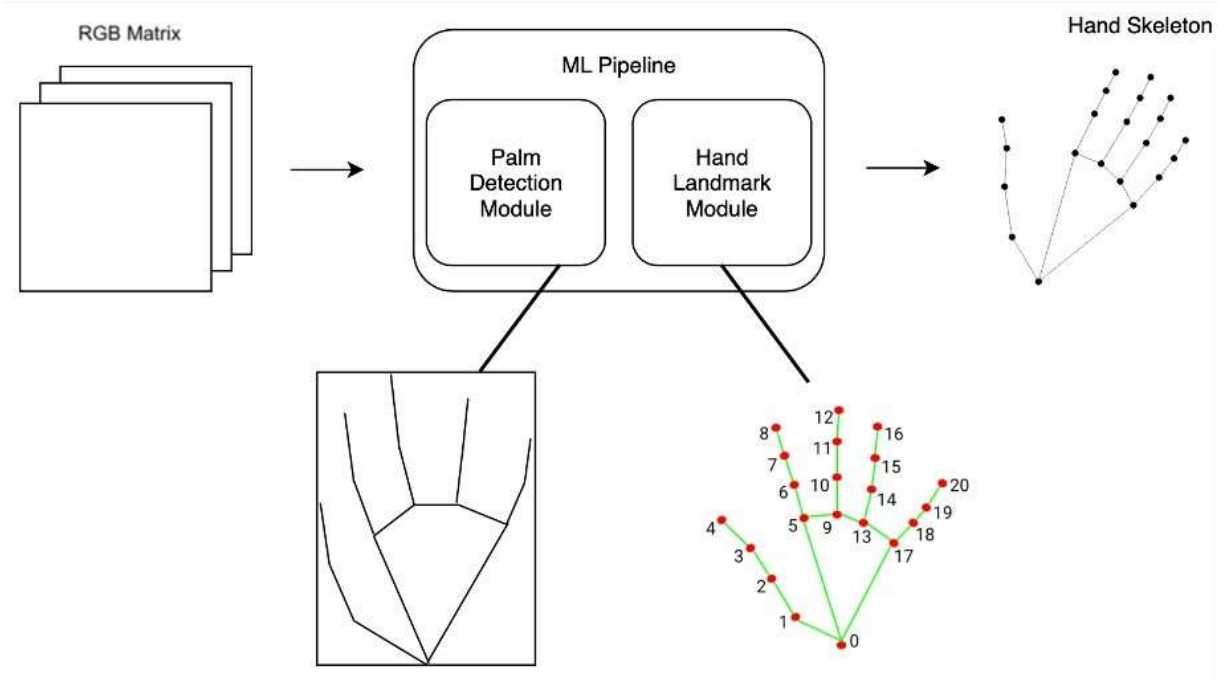

Fig. 2 Important Hand Landmarks

3.2 Facial Features Model using Haar Cascade Classifier

The facial feature detection component of our model utilizes a combination of traditional and deep learning techniques to accurately identify faces and analyze facial expressions in real-time video streams. Initially, we use a pre-trained Haar cascade classifier to detect faces within the video frames, leveraging characteristic patterns that indicate human faces. Detected faces are further analyzed using a convolutional neural network (CNN) trained specifically for facial emotion recognition, providing predictions such as ”Happy”, ”Sad” or ”Neutral”. Additionally, MediaPipe’s Holistic model complements face detection accuracy by detecting facial keypoints using its complete pose estimation feature. This enhances the drawing of annotations on the video frames to visualize the detected facial features and overall pose of the human face. By integrating these techniques sequentially, our model achieves robust face detection capabilities, enabling the evaluation of facial expressions in various real-world scenarios with high precision.

3.3 Data Acquisition and Creation

Our model captures 30-frame videos for each event sequence and create custom datasets in video form to better understand the hand signs. In this process, we use the MediaPipe Holistic model to extract keypoints from each frame, which are then flattened and assembled into NumPy arrays. These arrays are used to train the LSTM model, along with their corresponding event labels stored in lists. For sensitivity detection, the FER2013 dataset proves to be extremely valuable, providing nearly 29,000 RGB images of labeled faces across seven different emotional states. The users are also offered the option of creating their own custom dataset by adopting a webcam based approach to collect data. Each video captures 30 frames, focusing on recognizing common sign language actions such as ”hello”, ”good”, ”morning” and ”thank you”. Hand landmarks extracted from each frame contribute to a specialized dataset tailored for training our model to recognize these sign language gestures.

3.4 Extraction of keypoints

Our model uses the open-source framework called MediaPipe to precisely capture key palm keypoints that are crucial for interpreting sign language gestures. Instead of just detecting static signs, we focus on detecting full gestures through our model. MediaPipe’s palm detection model tracks the user’s palm movement, extracting key points with each frame. Each video captures 30 frames, allowing the entire gesture to be easily detected. These points are organized into a NumPy array, which then serves as input for training the LSTM model, enabling effective learning of complex palm gestures. Furthermore, the seamless integration of a face detection model within MediaPipe enables accurate identification and isolation of facial regions of interest (48x48 pixels) such as the eyes, nose, mouth, and overall face structure. This integrated approach ensures a thorough analysis, enabling the model to interpret both hand gestures and facial expressions accurately

3.5 Deep Learning Models

In predicting sign language gestures, we employ an LSTM (Long Short-Term Memory) model designed to process temporal sequences of gesture data. It has many distinct layers to give out the complex patterns and relationships within sign language. The model architecture consists of an LSTM layer with 64 units, ReLU activation, and it is configured to return sequences. Then, the second LSTM layer will contain 128 units, again configured to return sequences. A final LSTM with 64 units does not return sequences, effectively summarizing the learnt information. After these LSTM layers, it has two dense layers with 64 and 32 units respectively. All of these make use of the ReLU activation function to add non-linearity and enhance the learning capabilities. Finally, there is an output layer with a softmax activation function classifying gestures into predefined categories. The model is then compiled with the Adam optimizer, where categorical cross-entropy is utilized as a loss function, and it is trained over 2000 epochs to ensure robust performance. MediaPipe library is used for time hand gesture detection and landmark extraction to capture gestures and, in turn, key point extraction. We process the input frame, converting to RGB for the model to make a prediction, then it reverts back to BGR. Combining LSTM and Dense layers enables our model to manage the complexity and variability of sign language gestures, providing accurate and real-time predictions. The LSTM layers handle

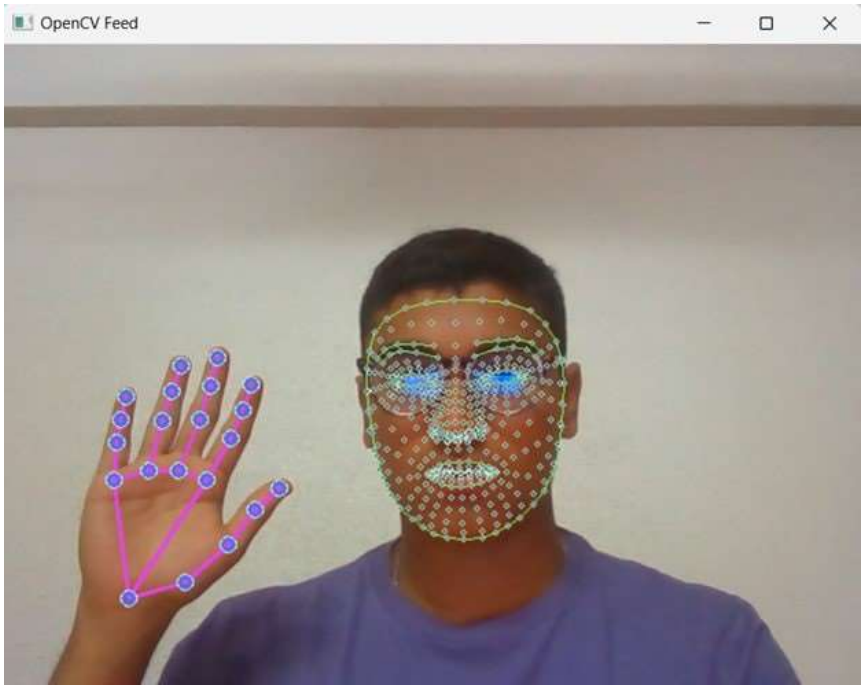

Fig. 3 Extraction of key points for sign language

the sequence dynamics, while the Dense layers fine-tune features for classification. MediaPipe enhances this by providing precise landmark detection and visualization.

3.6 Emotion Detection

Emotion recognition component of our system uses Open CV techniques with deep learning methodologies to analyse and interpret facial expressions from the face detected in every frame.The model is based around fine-tuned convolutional neural network (CNN) which is meticulously trained on large facial emotion datasets. The face is detected in a frame of the real time video by the Haar Cascade classifier. Detected faces are then cropped and resized to a fixed size of 48x48 pixels and feed into the CNN model which accurately predict emotions such as happiness, sadness, anger, and more.

With the use of complex neural networks, the CNN model analyses facial features, creating a probability distribution of emotions for each detected face. The real time video feed is then overlaid with the label of the most likely predicted emotion. Haar Cascade classifier continuously recognizes faces and refines the regions of interest, ensuring that complex emotions are detected even in dynamic environments. It also maintains a history of detected emotions by collecting predictions from each frame providing valuable insights.

Fig. 4 Emotion Detection Model

3.7 Merging Models

Facial expression and sign language recognition in real-time video streams is a radical development in human-computer interaction. The MediaPipe framework provides support for detailed interpretation of sign language and facial expressions, capturing subtleties of emotion. Through an amount of meticulous dataset creation through video image processing, the model reads hand and facial landmarks, mirroring the intricate details of the human hand and face. The hand gesture recognizer, developed using an LSTM neural network model architecture, goes beyond static signs and markers to interpret dynamic gestures, observing sequences of various hand formations over multiple frames. Concurrently, a CNN-based facial expression recognition model analyzes facial landmarks to categorize a variety of emotional states, capturing the essence of character sentiment with remarkable precision. An integrated approach would be able to evaluate sign language and emotional expressions in real time, making this tool very sophisticated for a variety of applications. It will not only provide instant sign language interpretation but will represent advanced inclusive and empathetic technology in communication.

3.8 Correlation of the models

The system's core processes the sequences of hand gestures to predict their meanings pertaining to sign language. In parallel, there will be a module for sentiment analysis on the emotional content of the recognized gestures. This is going to be achieved through semantic libraries like spaCy, leveraging meaningful sentences derived from the recognized gestures and handling grammatical correctness. This thus forms the basis for semantic understanding. Sentiment Analysis: This is performed with an NLTK Sentiment Intensity Analyzer, which will take these sentences and classify them as positive,

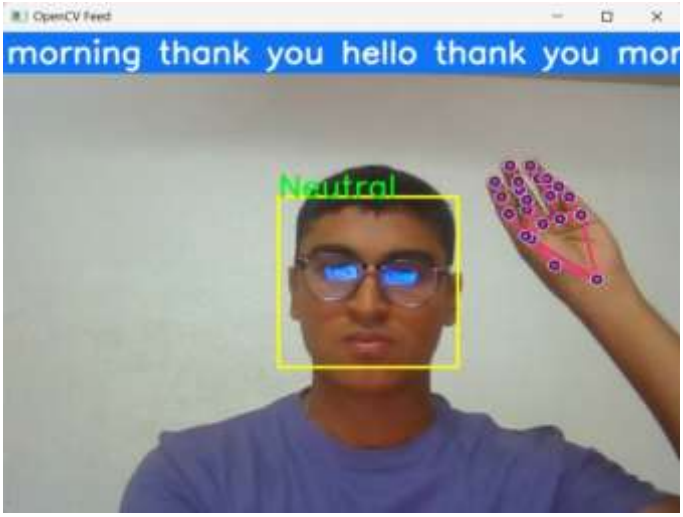

Fig. 5 Demonstration of how a sentence can be formed with the help of recognized words

negative, or neutral emotions, thus giving insight into the tone of the emotional communication. This can be necessary since the sentiment scores derived from the gesture analysis may mismatch the conveyed gestures and the user's real emotional state. This would therefore aid in identification and addressing inconsistencies where a user's gestures may suggest one emotional state while their facial expressions convey another. This may occur due to a number of reasons, including suppression of emotions or misinterpretation of gestures. This feature improves the reliability of communication by ensuring the emotional content of the sign language, matching the user's facial expressions. The system's output will also contain the recognized gesture and the relevant emotional state of the user, giving a full view of the communication process. Thus, by incorporating sentiment analysis with facial emotion data, the system ensures that the proper emotions which the user intends to convey are imbued in the conveyed message, therefore guaranteeing the empathy and contextual awareness of the communication tool and meeting the important requirement of consistency between gestures and emotional states.

```
Detected words: hello good morning thank you
Meaningful sentences: hello good morning thank you
Sentiment scores: [0.6597]
Overall sentiment score: 0.66
Overall facial emotion: Happy
Comparison: Sentiment vs Facial Emotion
Sentiment and facial emotion match.
```

Fig. 6 Correlation analysis

3.9 Audio Integration

One of the most crucial components of enhancing accessibility and user engagement is the text-to-speech (TTS) functionality. This feature makes it possible to convert text into generated speech, allowing users to interact with information through their sense of hearing. A set of operations controls the functionality of the TTS system. To maintain user control, the system manages global variables to adjust language preferences and handle program termination. After the whole sentence is translated by the core module , the translated audio is generated. This feature synthesizes voice asynchronously and interprets the input text using external libraries and modules like "gtts" and "translate." After translation, memory-handling techniques are used to integrate the synthesized speech into an audio stream. Multiple languages can be selected in the same session through threading. The speech generated is remembered by the system which leads to a more inclusive and accessible user experience.

Fig. 7 Text to Speech feature

4 Results

The proposed solution entails providing users dependent on sign language as a form of communication with sign Indian Sign Language (ISL) interpretation and emotion analysis. Using LSTM models and keypoint extraction techniques, our system interprets a wide range of sign language gestures, ensuring accurate and precise communication. This foundation forms the backbone of our model, helping in uninterrupted interactions for individuals within the sign language community.Additionally, our model integrates facial emotion recognition, providing insight into the emotional cues expressed during the instance when a sign is detected. By merging these components, our models adapt and respond based on both the conveyed message and the user's emotional expression, improving the interaction experience. Through attentive correlation analysis, our system ensures consistent communication, as showcased in Figures 8 and 9. In Figure 8 we are showing the first word (good) and the detected emotion (happy). In Figure 9, we show the instance of the sentence which is formed after making multiple signs (morning thank you hello thank you).
The accuracy of our LSTM Model for sign prediction on test data: 98.1228070%
The accuracy of our CNN Model for emotion detection on test data: 79.22%

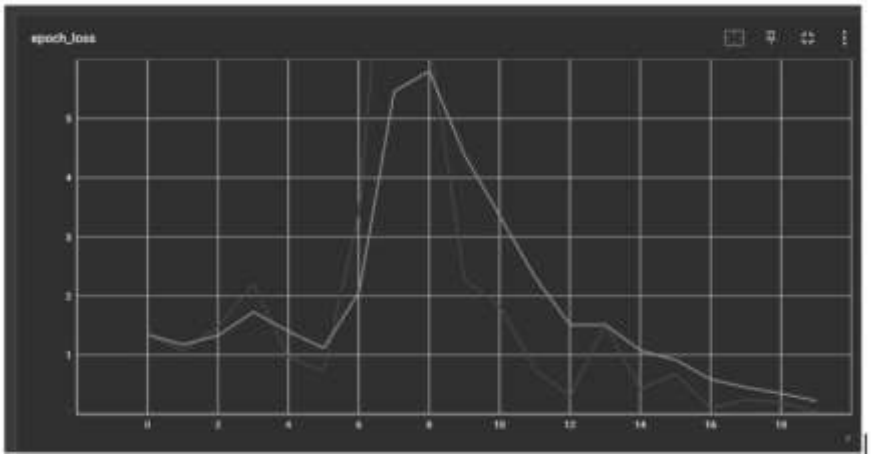

Fig. 8 Epoch loss during Training of Sign Language model

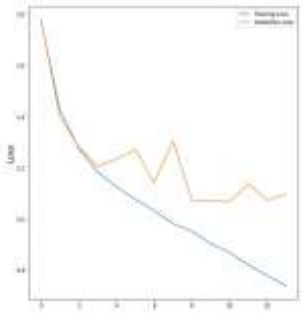

Fig. 9 Epoch loss during Training of Emotional recognition model

5 Comparison with other models

Papers	SL	Methods	Dataset	Size	Tr:Tt	Acc. (%)
[9] (2020)	ASL	RF, NB, SVM, LR, KNN, MLP	A-Z, 3-Spec. Char.	3000	75:25	72.23 - 98.31
[10] (2018)	ASL	Inception, LSTM	100 signs	2400	75:25	93.00
[1] (2020)	ASL	VGG-16	A-H	5000	75:25	99.62
[11] (2020)	ASL	SVM	A-Z, 'space'	3000	80:20	98.89
[12] (2020)	ISL	KNN	A-Z	220 DH, 800 SH	NA	95.84 DH, 94.88 SH
[13] (2014)	ISL	Haar Cascade	5 phrases	1000	90:10	92.68
Note: NA - Not Available, Tr:Tt - Training: Testing, SL - Sign Language						

6 Conclusion

Finally, the LSTM combined with CNN are a groundbreaking solution to recognize the challenges of Indian Sign Language which can be used for various applications. The system gives real time analysis and it is so valuable to hearing impaired individuals since communication barriers between the hearing and deaf people are broken by this system. Hence, this system strengthens the inclusivity of sign language users with the verbal communicators. The smart system with its ability to correctly interpret sign language gestures as well as the context of emotions to such an extent, not only improves the quality of communication, it also promotes better understanding regardless of the context.

The integration of advanced machine learning algorithms is also creating new applications in domains varying from everyday life to scientific research. Thus, the "translation" of sign language along with the emotional cues adds another dimension in addition to information reception and transmission. Evidence of this is that hearing loss can become a transformative force, helping individuals with hearing impairments integrate into an inclusive society.

7 Future Scope

Considering the future of this research, it would be of great significance to have user-centered mobile App, which will have the ability to accurate track the needs of individuals experiencing hearing loss at any place irrespective of their location. The app will have a feature that goes beyond providing comfort in communication but also to ensure that everyone is united and connected among themselves.

Furthermore, the development of a mobile application would also be a great way of increasing the reach and the usage of the app by being available on different digital platforms like desktops and tablets, through this our communication solution would impact lives positively. We seek to achieve this by plugging communication gaps, developing autonomy in individuals and enhancing the inclusiveness of the society.

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